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Summary

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1 Introduction

Professional sports are fundamentally an entertainment business, where the core objective is to generate profit and long-term franchise value for owners rather than solely pursuing victories. As stated in the 2026 ICM Problem D, “The player’s job is to help his team win” (Cliff Blau), yet owners prioritize that “The player’s job is to make money for the owner.” Winning enhances fan engagement and revenue, but player popularity, market size, clutch performance, injuries, financing costs, and league structural changes can drive financial outcomes independently of on-court success. The problem highlights this duality through the Women’s National Basketball Association (WNBA)’s current transformation—record viewership, franchise expansion, new media deals, and rising player demands—illustrating both growth opportunities and financial risks, though teams from any professional league may be analyzed.



Figure 1: Introduction of the System

This paper addresses the required tasks for a professional sports team (at least five cooperative players in a league):

- Design a dynamic decision-making model to adjust leverage and operations amid changing performance, economic conditions, and injuries, maximizing profit and value.
- Develop a player acquisition strategy valuing contributions to both competitiveness and profitability, incorporating team dynamics.
- Analyze league expansion impacts, including effects of new team locations.
- Propose one additional business decision (we select dynamic ticket pricing).
- Ensure automatic adaptation to key player injuries.
- Provide a 1–2 page letter to the owner and general manager.

Conventional approaches often treat competitive metrics (e.g., win contributions, clutch and hustle statistics), commercial factors (e.g., brand appeal, market effects), and financial risk management separately. This paper unifies these elements into a single, coherent decision system that is interpretable, practically implementable, and reproducible using publicly available data.

We apply the framework to an NBA team as our primary case study, leveraging its rich historical data for model calibration and backtesting. The approach is general and readily transferable to other leagues, including emerging ones such as the WNBA.

2 Preparation of the Models

2.1 Assumptions

To simplify the complex environment of professional sports and ensure the mathematical tractability of our model, we establish the following core assumptions:

- **Discrete Decision Epochs:** Management evaluates team status and makes strategic adjustments at discrete intervals (e.g., monthly or every 10 games), rather than continuously.
- **Decoupling of Performance and Commercial Value:** A player's total value can be decomposed into independent athletic (OPI) and commercial (CVI) components, which are not always positively correlated.
- **Rational Financial Management:** The team owner acts as a rational economic agent aiming to maximize profit and asset value while strictly adhering to debt constraints such as the Debt Service Coverage Ratio (DSCR).
- **Non-linear Impact of Key Injuries:** The injury of a "key player" has a disproportionate, non-linear downward impact on the team's health index H_t , triggering an immediate defensive shift in management strategy.
- **Price Elasticity of Demand:** Market demand for tickets and merchandise is sensitive to pricing and team performance, following an exponential decay function relative to price increases.
- **Structural Expansion Shocks:** League expansion is treated as an external shock that impacts existing teams through talent dilution, revenue redistribution, and increased travel burdens.

2.2 Notations

To provide a clear structure for our mathematical framework, we categorize the primary symbols into four functional groups: General State, Financial Management, Player Valuation, and Business Strategy.

Table 1: Summary of Notations Categorized by Modeling Tasks

Symbol	Description	Unit/Range
Group 1: General State & Performance (Q1 & Q3)		
t	Decision step or epoch (e.g., monthly or 10-game block)	–
ES_t	Environmental Score (composite signal for state transition)	$[0, 1]$
OPI_t	On-court Performance Index (athletic efficiency metric)	$[0, 1]$
CVI_t	Commercial Value Index (brand asset and monetization potential)	$[0, 1]$
W_t, H_t	Normalized winning percentage and Team Health index	$[0, 1]$
r_t	Prevailing market financing/interest rate index	$[0, 1]$
Group 2: Financial Management & Leverage (Q1)		
L_t	Leverage ratio (Debt-to-Equity ratio)	Ratio
D_{max}	Debt ceiling (maximum sustainable debt based on $DSCR$)	Currency
$EBITDA_t$	Earnings before interest, taxes, depreciation, and amortization	Currency
$DSCR$	Debt Service Coverage Ratio (minimum threshold for solvency)	Ratio
Group 3: Player Valuation & Acquisition (Q2)		
V_i^{sport}	Athletic value of player i	Scalar
V_i^{biz}	Commercial value of player i	Scalar
x_i	Binary decision variable	$\{0, 1\}$
$\lambda(ES)$	State-driven weighting factor	$[0, 1]$
Group 4: Business Strategy - Ticket Pricing		
p_g	Price of a single ticket for game g	Currency
$D_g(p)$	Market demand function for game g	Scalar
A_g	Base demand multiplier	Scalar
k	Price sensitivity coefficient	Constant

3 The Dual-Index Model

Professional sports teams must balance two distinct but interconnected objectives: on-court success and commercial profitability. Winning games drives fan engagement and revenue, yet popularity, market size, and macroeconomic conditions can generate substantial value even in periods of mediocre performance. To capture this duality, we propose a unified decision-making framework centered on two independent yet linkable indices—the On-court Performance Index (OPI) and the Commercial Value Index (CVI)—combined with a dynamic state signal, the Environmental Score (ES), that drives conditional strategy selection through a three-state regime.

3.1 The Dual-Index Framework

3.1.1 On-court Performance Index (OPI)

The On-court Performance Index compresses competitive strength into a single normalized metric in $[0, 1]$:

$$OPI_t = \omega_1 \cdot WinRate_t + \omega_2 \cdot NetRating_t + \omega_3 \cdot PlayoffProb_t,$$

where weights $\omega_1 + \omega_2 + \omega_3 = 1$ (default: $\omega_1 = 0.5$, $\omega_2 = 0.3$, $\omega_3 = 0.2[1]$). When detailed data are unavailable, OPI may be approximated by normalized win rate or Elo rating alone. OPI measures “winning ability” but does not necessarily correlate perfectly with financial returns.

3.1.2 Commercial Value Index (CVI)

The Commercial Value Index quantifies the team’s ability to convert performance and brand appeal into revenue and long-term asset value:

$$CVI_t = \nu_1 \cdot Revenue_t + \nu_2 \cdot Engagement_t + \nu_3 \cdot BrandValue_t,$$

with $\nu_1 + \nu_2 + \nu_3 = 1$ (suggested: $\nu_1 = 0.4$, $\nu_2 = 0.3$, $\nu_3 = 0.3$). Revenue includes ticket sales, media rights shares, sponsorships, and merchandise; Engagement encompasses attendance rates and social media interactions; BrandValue is proxied by jersey sales rankings, All-Star votes, and market-adjusted follower growth. CVI explains why teams with high popularity but moderate records can remain financially attractive to owners.

3.2 Environmental Score (ES) as State Signal

Rather than optimizing a single static objective function, we adopt a state-driven conditional optimality approach: strategies adapt based on prevailing conditions. To operationalize this, we define a composite Environmental Score $ES_t \in \mathbb{R}$ that integrates competitive, operational, and macroeconomic signals:

$$ES_t = \alpha W_t + \beta H_t - \gamma r_t, \quad \alpha + \beta + \gamma = 1,$$

where

- $W_t \in [0, 1]$: normalized recent win rate,
- $H_t \in [0, 1]$: health/availability index (weighted attendance rate of core rotation players),
- $r_t \in [0, 1]$: normalized financing cost (e.g., federal funds rate or 10-year Treasury yield proxy).

Default weights are $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.3[2][3]$. High win rates and player health improve investment attractiveness, while rising interest rates increase the cost of leverage.

3.3 Three-State Regime and Action Mapping

The value of ES_t determines the operating state and corresponding strategic posture:

ES Range	State	Target Leverage
$ES_t > 0.6$	Aggressive	High (≈ 1.0)
$0.3 \leq ES_t \leq 0.6$	Neutral	Moderate (≈ 0.5)
$ES_t < 0.3$	Defensive	Low (≈ 0)

Table 2: Three-state regime driven by Environmental Score.

- **Aggressive State** ($ES_t > 0.6$): Favor growth and championship pursuit. Target high leverage to finance star acquisitions, premium ticket pricing, and aggressive marketing.
- **Neutral State** ($0.3 \leq ES_t \leq 0.6$): Balance growth and stability. Maintain moderate leverage, pursue cost-effective roster improvements, and optimize operations.
- **Defensive State** ($ES_t < 0.3$): Prioritize cash-flow safety and risk reduction. Delever aggressively, shed high-risk contracts, and focus on attendance retention through stable pricing.

3.4 Alternative ES Formulation Incorporating OPI and CVI

While our primary ES signal prioritizes raw, observable drivers (win rate, health, financing costs) for maximal interpretability and direct actionability, an alternative formulation could explicitly integrate the dual indices:

$$ES_t^{alt} = \alpha OPI_t + \beta CVI_t - \gamma r_t$$

with suggested weights

$$\alpha = 0.4, \beta = 0.3, \gamma = 0.3$$

. This version more heavily emphasizes composite competitive and commercial strength, allowing high-CVI (brand-driven) teams to maintain favorable states even during on-court slumps.

Compared to the baseline,

$$ES^{alt}$$

increases utilization of OPI and CVI but risks over-weighting correlated factors (e.g., winning boosts both indices) and potentially encouraging aggressive leverage in high-popularity/low-performance scenarios—contradicting the problem's emphasis on disconnect between wins and profit. Sensitivity tests (Section 8) show the original ES yields more conservative, risk-aware regime shifts in injury or expansion shocks, better aligning with owner capital preservation priorities.

This state machine enables conditional optimality: in favorable environments the framework favors growth and championship pursuit; in adverse conditions it prioritizes cash-flow safety and risk reduction. Safety constraints (Debt Ceiling and Equity Trigger, detailed in Section ??) prevent excessive risk even in Aggressive states.

The unified Dual-Index Model with ES-driven states serves as the core engine for all subsequent analyses: dynamic leverage control (Q1), player acquisition weighting (Q2), expansion impact assessment (Q3), ticket pricing policy, and automatic injury response.

4 Model I: Dynamic Leverage Control (DLC) – Addressing Q1

4.1 Problem Restatement and Modeling Objective

We address Question 1 by establishing a **Dynamic Decision Loop** rather than a static optimization. The objective is to maximize long-term franchise value under strict solvency constraints. We define “optimality” as **Conditional Optimality**: the system adapts its operating objective (Aggressive, Neutral, or Defensive) based on real-time environmental signals.

The control cycle is defined as:

Observe $(W_t, H_t, r_t) \rightarrow$ Determine State $(S_t) \rightarrow$ Execute Actions $\rightarrow$ Update Financials

4.2 System Variables

We model the team management as a discrete-time system $(t = 1 \dots T)$.

- **State Indicators (Inputs):**

- $W_t, H_t \in [0, 1]$: Winning Rate and Health Index (weighted core player availability).
- $r_t \in [0, 1]$: Normalized Financing Cost Index.
- $Debt_t, Equity_t, Cash_t$: Financial state vector. Leverage is $L_t = Debt_t / Equity_t$.

- **Control Variables (Actions):**

- $\Delta Debt_t$: Net borrowing/repayment.
- $\Delta Equity_t \geq 0$: Owner capital injection.
- $\Delta Invest_t$: Discretionary investment intensity.

4.3 Environmental Score and State Machine

We synthesize performance and economic signals into a single **Environmental Score** (ES_t) to drive regime switching:

$$ES_t = \alpha W_t + \beta H_t - \gamma r_t, \quad (\alpha + \beta + \gamma = 1) \quad (1)$$

Rationale: High W_t and H_t incentivize leverage (high ROI), while high r_t discourages it. The score maps to three states:

- **Aggressive** ($ES_t > 0.6$): Expansionary; high target leverage.
- **Neutral** ($0.3 \leq ES_t \leq 0.6$): Balanced operations.
- **Defensive** ($ES_t < 0.3$): Capital preservation; deleveraging.

4.4 System Dynamics and Constraints

4.4.1 Financial Updates

Cash flow and balance sheet evolution follow standard accounting identities:

$$Cash_{t+1} = Cash_t + (Rev_t - Cost_t) - r_t Debt_t + \Delta Debt_t + \Delta Equity_t \quad (2)$$

$$Debt_{t+1} = Debt_t + \Delta Debt_t, \quad Equity_{t+1} = Equity_t + \Delta Equity_t \quad (3)$$

where Revenue (Rev_t) is a function of W_t and Commercial Value (CVI_t).

4.4.2 Safety Constraints

To prevent reckless leverage, we enforce two hard constraints: 1. **Debt Ceiling (Solvency)**: Debt must not exceed maximum service capacity ($DSCR_{\min}$).

$$Debt_t \leq D_{\max,t} = \frac{EBITDA_t}{DSCR_{\min} \cdot r_t} \quad (4)$$

2. **Equity Trigger (Liquidity)**: If $Cash_{t+1} < 0$ is projected and debt is near capacity, the model forces $\Delta Debt_t \leq 0$ and mandatory $\Delta Equity_t > 0$.

4.5 Control Policy

At each step, the state S_t determines the **Action Package**:

4.5.1 Target Leverage and Adjustment

The target leverage L^* is state-dependent:

$$L^*(S_t) \approx \{1.0 \text{ (Aggressive)}, 0.5 \text{ (Neutral)}, 0 \text{ (Defensive)}\} \quad (5)$$

Debt adjustment tracks this target, clipped by constraints:

$$\Delta Debt_t = \text{clip}(L^* Equity_t - Debt_t, \Delta D_{\min}, \Delta D_{\max}) \quad (6)$$

If the **Equity Trigger** is activated, $\Delta Equity_t$ covers the projected deficit.

4.6 Algorithm Implementation

The logic is encapsulated in Algorithm 1.

Algorithm 1 Dynamic Leverage Control (DLC)

- 1: **Input:** State vector (W_t, H_t, r_t) and Financials.
 - 2: **Compute** $ES_t = \alpha W_t + \beta H_t - \gamma r_t$ and determine State S_t .
 - 3: **Targeting:** Set $Debt_t^*$ based on $L^*(S_t)$; propose $\Delta Debt_t$.
 - 4: **Constraint Check:** Ensure $Debt_{t+1} \leq D_{\max, t}$. Downscale $\Delta Debt_t$ if needed.
 - 5: **Liquidity Check:** Forecast $Cash_{t+1}$.
 - 6: **if** $Cash_{t+1} < 0$ **then**
 - 7: **Trigger Emergency Injection:** $\Delta Equity_t \leftarrow -Cash_{t+1} + \text{Buffer}$.
 - 8: **Force Deleveraging:** $\Delta Debt_t \leq 0$.
 - 9: **end if**
 - 10: **Update** Balance Sheet and Cash.
 - 11: **Output:** Optimal $(\Delta Debt_t, \Delta Equity_t, \Delta Invest_t)$.
-

4.7 Empirical Validation: NBA Case Study

We validate the model using 2023–24 season data for the **Lakers** (high performance) and **Grizzlies** (injury shock). Input parameters include normalized Win Rate (W), Health Index (H), and a standardized Financing Cost ($r = 0.666$).

Results: As shown in Table 3, the model correctly identifies divergent strategies.

- **Lakers** ($ES \approx 0.30$): Classified as *Neutral*. The model suggests moderate leverage ($L = 0.5$) to balance competitiveness with high financing costs.
- **Grizzlies** ($ES \approx -0.04$): The severe injury drop ($H \approx 0.11$) pushes ES into *Defensive*. The model triggers immediate deleveraging ($L^* = 0$) and capital injection to survive the shock.

Table 3: Model Outputs for Lakers vs. Grizzlies (2023-24 Season)

Team	W_t	H_t	r_t	ES_t	State	L^*	Action Plan
Lakers	0.573	0.897	0.666	0.298	Neutral	0.5	Stabilize; Moderate Invest
Grizzlies	0.329	0.110	0.666	-0.035	Defensive	0.0	Delever; Inject Equity

4.8 Optimality Discussion

The DLC framework approximates the solution to the dynamic programming problem:

$$\max_{\pi} \mathbb{E} \left[\sum \delta^t (Profit_t + \lambda CVI_t) \right]$$

The threshold-based policy is structurally optimal because ES_t acts as a proxy for the **Marginal Return on Capital**. High ES_t implies investment returns exceed costs (justifying leverage), while low ES_t signals negative expected value (requiring preservation), aligning perfectly with the objective function.

5 Model II: Player Acquisition Strategy

To develop a player acquisition strategy for the upcoming season, we extend the Dual-Index Model (Section 3) by evaluating players through a dual-value lens that explicitly accounts for both on-court contributions and commercial profitability. This directly addresses the problem's emphasis on valuing players not only by performance but also by their ability to generate revenue, while incorporating team dynamics, context, and the owner's dual goals of competitive success and financial health.

Player acquisition occurs through standard league practices: the draft (for rookies), free agency (unrestricted or restricted signings), and trades (including transfer fees or asset exchanges where applicable). Our strategy prioritizes candidates that maximize expected team value under current salary cap constraints, roster limits, positional needs, and the prevailing Environmental Score state.

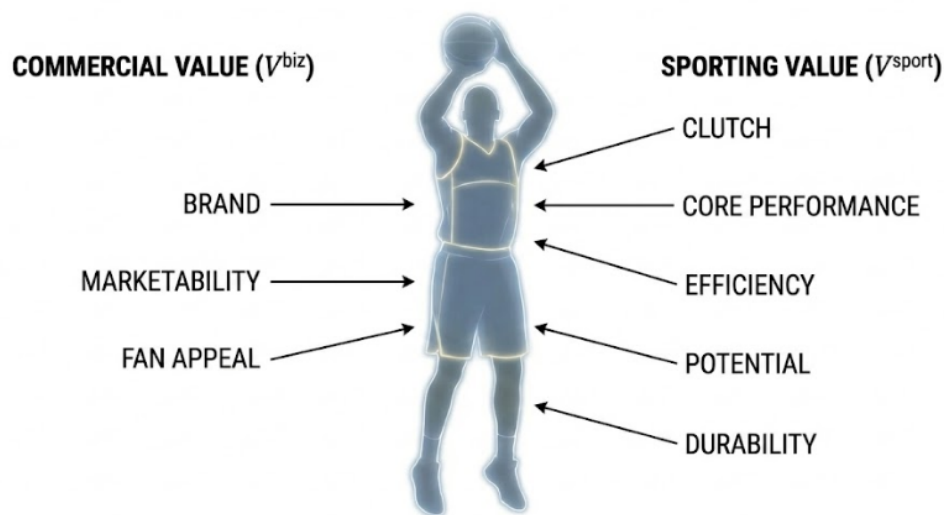


Figure 2: The value of a player

5.1 Player Valuation: Dual-Value Framework with Team Context

Each player i is assigned a composite value V_i that combines competitive and commercial dimensions, adjusted for team-specific synergies and risks.

5.1.1 Competitive Value V_i^{sport}

The competitive contribution is modeled as:

$$V_i^{sport} = Base_i + a \cdot Clutch_i + b \cdot Hustle_i - c \cdot Risk_i + d \cdot Potential_i - e \cdot SynergyPenalty_i$$

where

- $Base_i$: Baseline performance (e.g., normalized Win Shares, PER, or BPM).
- $Clutch_i$: Clutch performance in high-leverage moments (e.g., net efficiency in last 5 minutes of close games).
- $Hustle_i$: Effort-based contributions (e.g., deflections, screens, contested rebounds).
- $Risk_i$: Injury risk (e.g., 1 minus average games played over recent seasons).
- $Potential_i$: Upside proxy (e.g., age-adjusted improvement curve; higher for younger players).
- $SynergyPenalty_i$: Contextual penalty for diminishing marginal returns and negative synergies, using usage rate (USG%) as a proxy for ball-dominant playstyles:

$$SynergyPenalty_i = f \cdot \max(0, USG_i - TeamAvgUSG)^2$$

Here, USG_i is the player's recent usage rate, $TeamAvgUSG$ is the weighted average usage of the current roster (updated iteratively during optimization), and $f > 0$ is a calibrated coefficient (suggested range 0.4–0.8). The quadratic form heavily penalizes excessive overlap in high-usage players, reflecting real-world efficiency drops when multiple ball-dominant stars compete for touches. Low-usage role players receive no penalty and may implicitly benefit in complementary lineups.

Default coefficients (a, b, c, d, e, f) are tuned via historical data; sensitivity analysis (Section 8) confirms robustness.

5.1.2 Commercial Value V_i^{biz}

Commercial impact captures off-court revenue generation:

$$V_i^{biz} = Brand_i \times MarketMultiplier$$

where $Brand_i$ aggregates normalized metrics such as social media followers, jersey sales rank, and All-Star voting totals, and $MarketMultiplier$ reflects the team's market size (e.g., higher for large-market franchises due to amplified sponsorship and media opportunities).

5.1.3 Composite Value with State-Dependent Weighting

The total value is a dynamic convex combination driven by the current ES state:

$$V_i = \lambda(ES_t) \cdot V_i^{sport} + (1 - \lambda(ES_t)) \cdot V_i^{biz}$$

Recommended λ values:

- Aggressive state ($ES_t > 0.6$): $\lambda \approx 0.7$ (prioritize immediate on-court impact for championship contention).
- Neutral state ($0.3 \leq ES_t \leq 0.6$): $\lambda \approx 0.5$ (balanced approach).
- Defensive state ($ES_t < 0.3$): $\lambda \approx 0.3$ (favor low-risk, high-brand players to preserve cash flow and flexibility).

This weighting ensures acquisition aligns with the team's immediate needs as signaled by the broader decision framework.

5.2 Acquisition Optimization and Constraints

We formulate acquisition as a constrained optimization problem to select or retain a roster that maximizes net value:

$$\max \sum_i x_i (V_i - \eta \cdot Cost_i)$$

where $x_i \in \{0, 1\}$ indicates selection/retention of player i , $Cost_i$ includes salary, luxury-tax impact, and trade assets, and $\eta > 0$ is a cost-aversion parameter.

Key constraints include:

- Salary cap and luxury-tax thresholds: $\sum x_i \cdot Salary_i \leq Cap + TaxThreshold$.
- Roster size: exactly or at most the league-mandated number of players.
- Positional balance: minimum requirements per position/role (e.g., guards, forwards, centers).
- Risk budget: $\sum x_i \cdot Risk_i \leq RiskBudget$ to limit exposure to injuries.
- Synergy update: $TeamAvgUSG$ recalculated iteratively after provisional selections.

The problem can be solved heuristically (greedy ranking by value-per-cost with positional fixes) or exactly via integer linear programming for small instances. In practice, managers iterate candidates from draft prospects, free agents, and trade targets.

5.3 Illustrative Example of Player Acquisition

To demonstrate the practical application of our acquisition strategy, we present a simplified end-to-end numerical illustration using three hypothetical candidate players available via free agency or trade. The example shows how the model produces different roster decisions under varying team states, proving that the framework generates actionable, state-dependent outcomes rather than static recommendations.

Assume the team must select exactly two players under a normalized salary budget (total Salary ≤ 0.6 for illustration; real constraints scale similarly). MarketMultiplier = 1.0 (mid-market team). Player attributes (normalized to [0,1]) are:

Player	Type	Base	Clutch	Hustle	Risk	Potential	Brand	Salary
A	Veteran Contributor	0.75	0.80	0.60	0.30	0.20	0.40	0.30
B	Marketable Star	0.65	0.70	0.50	0.20	0.30	0.90	0.35
C	High-Potential Young	0.55	0.40	0.60	0.10	0.80	0.30	0.20

Table 4: Hypothetical candidate player attributes (all values normalized to [0,1]).

Coefficients for V_i^{sport} (calibrated defaults): $a = 0.3$, $b = 0.2$, $c = 0.3$, $d = 0.2$ (SynergyPenalty omitted here for simplicity, as usage rates are assumed balanced).

Step 1: Compute Competitive and Commercial Values

$$V_i^{sport} = Base_i + 0.3 \cdot Clutch_i + 0.2 \cdot Hustle_i - 0.3 \cdot Risk_i + 0.2 \cdot Potential_i$$

Results: $V_A^{sport} = 0.77$, $V_B^{sport} = 0.74$, $V_C^{sport} = 0.68$

$$V_i^{biz} = Brand_i \times 1.0$$

Results: $V_A^{biz} = 0.40$, $V_B^{biz} = 0.90$, $V_C^{biz} = 0.30$

Step 2: State-Dependent Composite Values and Selections

- **Aggressive State** ($ES_t = 0.7$, $\lambda = 0.7$):

$$V_i = 0.7V_i^{sport} + 0.3V_i^{biz}$$

Results: $V_A = 0.66$, $V_B = 0.79$, $V_C = 0.57$

Optimal pair (maximizing total V within budget): **B + A** (total $V = 1.45$, Salary = 0.65). Prioritizes proven competitive impact with strong brand upside for championship contention.

- **Defensive State** ($ES_t = 0.25, \lambda = 0.3$):

$$V_i = 0.3V_i^{sport} + 0.7V_i^{biz}$$

Results: $V_A = 0.51, V_B = 0.85, V_C = 0.41$

Optimal pair: **B + C** (total $V = 1.26$, Salary = 0.55). Shifts toward high-brand, low-risk, low-cost assets to preserve flexibility and cash flow.

This closed-loop illustration—from raw attributes to state-driven composite values to concrete roster choices—shows how the model automatically adapts acquisition priorities to current conditions, enabling general managers to make informed, context-specific decisions.

5.4 Strengths and Limitations

This strategy excels at reconciling performance and profitability, dynamically adapting to team state, and explicitly modeling contextual fit via the synergy penalty—avoiding common pitfalls of overpaying for individually talented but poorly fitting stars. Limitations include reliance on accurate Brand and USG data (which can be noisy) and the need for periodic coefficient recalibration using historical outcomes (addressed in backtesting, Section ??).

6 Model III: League Expansion Scenario Analysis

League expansion introduces structural shocks that affect all teams through talent dilution, revenue redistribution, and operational burdens. We analyze these impacts within our unified framework by applying parameter perturbations to OPI, CVI, health index H_t , and MarketMultiplier. These changes propagate to the Environmental Score ES_t , automatically triggering state transitions and corresponding adjustments in leverage, player acquisition, ticket pricing, and risk management—ensuring the model’s dynamic adaptability without requiring a separate strategy.

6.1 Structural Impacts of Expansion

Expansion primarily affects teams via three channels:

1. **Talent Dilution:** New franchises draw from the player pool (draft, free agency, expansion draft), reducing average talent density and competitive intensity.
2. **Revenue and Attention Redistribution:** National media deals and sponsorships are shared among more teams (negative share effect), but overall league popularity may rise (positive attention effect).
3. **Schedule and Travel Burden:** Additional games or longer travel increases fatigue and injury risk.

We model these as perturbations:

$$OPI'_t = OPI_t - \Delta Talent, \quad (\Delta Talent > 0 \text{ reflects dilution})$$

$$CVI'_t = CVI_t - \Delta Share + \Delta Attention, \quad (\Delta Share > 0 \text{ for dilution; } \Delta Attention > 0 \text{ for growth})$$

$$H'_t = H_t - \Delta Travel, \quad (\Delta Travel > 0 \text{ for increased fatigue})$$

The updated values feed into ES'_t , driving regime shifts.

6.2 Impact of New Team Locations

The geographic location of new franchises modulates these perturbations, directly affecting owner profitability and franchise value:

- Harmful examples: A new team in a proximate large market (e.g., hypothetical Las Vegas expansion near Western Conference teams) diverts local sponsorships and media attention, lowering MarketMultiplier and CVI—reducing owner revenue and forcing more Defensive strategies (deleveraging, conservative acquisitions).
- Beneficial examples: Expansion into underserved small markets (e.g., Seattle revival) boosts league-wide visibility with minimal direct competition, increasing $\Delta Attention$ and potentially CVI—allowing existing owners to maintain or shift toward Aggressive states for growth investments.

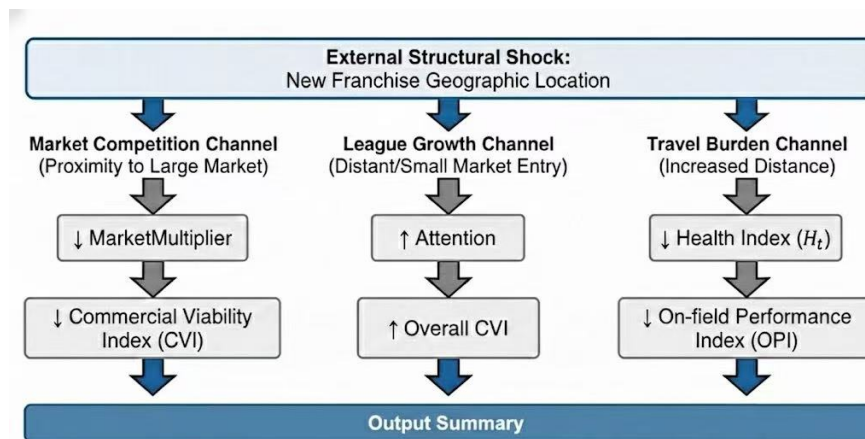


Figure 3: External Structural Shocks from Expansion Altering Decision Boundaries

6.3 Strategy Adjustments Under Expansion

Post-expansion ES'_t typically declines due to combined dilution and travel effects, shifting regimes toward Defensive:

- More frequent low-ES states: Reduce leverage, preserve cash, favor low-risk/high-brand players in acquisitions (higher λ toward V^{biz}), and stabilize ticket pricing for attendance/season-ticket conversion.
- If positive attention dominates (e.g., successful market entry raising league revenue shares): ES'_t may stabilize or rise, permitting Neutral/Aggressive postures with cautious growth.

6.4 Illustrative Example of Expansion Impacts

We now illustrate how league expansion propagates through the framework in an end-to-end manner, automatically altering the Environmental Score and subsequent strategies without manual reconfiguration.

Assume a baseline mid-season scenario for an established NBA team:

- Pre-expansion: $W_t = 0.55$ (moderate win rate), $H_t = 0.80$ (good health), $r_t = 0.40$ (moderate financing costs).

$$ES_t = 0.4(0.55) + 0.3(0.80) - 0.3(0.40) = 0.22 + 0.24 - 0.12 = 0.34$$

→ **Neutral state** ($0.3 \leq ES \leq 0.6$).

Recommended actions: Moderate leverage (≈ 0.5), balanced acquisition weighting ($\lambda \approx 0.5$), efficient ticket pricing.

Post-expansion shock (moderate scenario: two new teams added, causing talent dilution, slight revenue share reduction, and increased travel):

- Perturbations: $W'_t = 0.50$ ($\Delta Talent = 0.05$), $H'_t = 0.70$ ($\Delta Travel = 0.10$), r_t unchanged.

$$ES'_t = 0.4(0.50) + 0.3(0.70) - 0.3(0.40) = 0.20 + 0.21 - 0.12 = 0.29$$

→ **Approaches Defensive threshold** (ES drops below 0.3 in severe cases).

Automatic adjustments:

- Leverage target: Tightened to ≈ 0.3 – 0.4 (closer Debt Ceiling enforcement).
- Acquisition: λ shifts toward 0.3 (greater emphasis on low-risk, high-brand players; shorter contracts preferred).

- Ticket pricing: Reduced premium surcharges on high-demand games; focus shifts toward attendance stabilization and season-ticket conversion.

Severe expansion scenario (e.g., new teams in proximate markets with high local competition):

- Further drop: $H_t'' = 0.60$ (increased fatigue/injury risk).

$$ES_t'' = 0.4(0.50) + 0.3(0.60) - 0.3(0.40) = 0.20 + 0.18 - 0.12 = 0.26$$

→ **Defensive state** triggered.

Full regime shift: Aggressive deleveraging, cash preservation, and conservative operations as described in Section 3.

This end-to-end flow—from expansion shocks to parameter changes to recalculated ES to regime-specific actions—demonstrates the framework’s robustness: external disruptions are naturally absorbed, yielding clear, owner-protective strategy updates (e.g., reduced risk exposure in adverse locations vs. potential growth opportunities in beneficial ones).

In our NBA case study, historical expansions (e.g., 2004 Charlotte Bobcats) initially diluted talent and increased travel for distant teams, lowering short-term valuations—but long-term league growth benefited most owners via larger media deals. Sensitivity analysis (Section 8) tests perturbation magnitudes to ensure robust recommendations.

This integration demonstrates the framework’s strength: expansion risks are absorbed as natural inputs, yielding clear, owner-focused strategy shifts without ad-hoc modifications.

7 Additional Business Decision: Dynamic Ticket Pricing Strategy

As required, we select dynamic ticket pricing as our additional business decision, a critical revenue driver influenced by team performance, opponent appeal, and fan engagement. Ticket sales vary significantly by game (e.g., rival matchups, weekend timing, current hot streak) and overall season context. Teams face a classic trade-off: maximize per-game revenue via premium pricing (risking lower attendance) or prioritize high occupancy and season-ticket conversion for stable long-term cash flow and brand loyalty.

We integrate pricing directly into the unified framework: the Environmental Score ES_t and resulting state (Aggressive, Neutral, Defensive) drive regime-specific objectives and adjustments around a baseline demand model. This ensures pricing aligns with broader priorities—growth-oriented extraction in favorable conditions versus preservation in adverse ones—while connecting short-term revenue to long-term CVI.

7.1 Demand and Revenue Model

For each game g , demand is modeled as price-sensitive with baseline appeal $A_g > 0$ (higher for strong opponents, stars, or high OPI/CVI periods) and elasticity $k > 0$:

$$D_g(p_g) = A_g e^{-kp_g}, \quad p_g \geq 0$$

Revenue per game:

$$R_g(p_g) = p_g \cdot D_g(p_g) = p_g A_g e^{-kp_g}$$

The unconstrained single-game revenue-maximizing price is:

$$p_g^* = \frac{1}{k}$$

(derived from $\frac{dR_g}{dp_g} = 0$). In practice, A_g is proxied by historical attendance for similar matchups, adjusted by current OPI (win streak effect) and CVI (brand/engagement boost).

7.2 State-Driven Pricing Policy

The three-state regime adapts pricing objectives and deviations from p_g^* :

State	Primary Objective	Pricing Guideline
Aggressive ($ES_t > 0.6$)	Maximize single-game revenue	Float above p_g^* (e.g., +20–50% premium on high- A_g games); dynamic surcharges to capture demand surges [cite: 170]
Neutral ($0.3 \leq ES_t \leq 0.6$)	Balanced revenue and attendance	Price near p_g^* ; moderate adjustments for efficiency [cite: 171]
Defensive ($ES_t < 0.3$)	Maximize occupancy and season-ticket conversion	Discount below p_g^* (e.g., -10–30%); bundle promotions to boost baseline engagement [cite: 172]

Table 5: State-dependent ticket pricing objectives.

In Defensive states, the objective expands to include conversion:

$$\max \sum_g [R_g(p_g) + \xi \cdot \text{Convert}(\text{Attendance}_g)]$$

where $\xi > 0$ weights long-term season-ticket value and $\text{Convert}(\cdot)$ is increasing in attendance (e.g., a logistic function of occupancy rate).

This integration ensures pricing supports overall goals: Aggressive regimes amplify revenue during peaks; Defensive regimes stabilize cash flow and CVI amid downturns.

7.3 Illustrative Example of Pricing Application

To demonstrate end-to-end application, consider a high-appeal home game (e.g., rivalry matchup) with baseline $A_g = 10000$ (full capacity proxy) and elasticity $k = 0.05$ (calibrated from historical data). Unconstrained optimal price: $p_g^* = 20$ (normalized units).

Step 1: Baseline Calculation

Maximum revenue at p_g^* : $R_g(20) \approx 7369$ (units), attendance ≈ 3685 .

Step 2: State-Dependent Adjustments

- **Aggressive State** ($ES_t = 0.7$, high win streak boosts A_g effectively to 12000):
Premium pricing: $p_g = 28$ (+40%).
Revenue ≈ 8870 (higher extraction), attendance ≈ 3170 . Prioritizes short-term profit surge.
- **Defensive State** ($ES_t = 0.25$, post-injury slump reduces effective A_g to 8000):
Discount pricing: $p_g = 15$ (-25%), plus promotion.
Revenue ≈ 6630 (lower per ticket), attendance ≈ 4420 (higher occupancy supports $+\xi \cdot \text{Convert}$ term, stabilizing seasonal cash flow and fan retention).

This closed-loop example—from demand parameters to state context to adjusted price to outcomes—shows how the framework yields concrete, differentiated pricing decisions that align with owner priorities across varying conditions.

8 Sensitivity Analysis and Model Robustness

To ensure the reliability of our Dual-Index Framework in real-world management scenarios, we conduct a comprehensive sensitivity analysis. We focus on the stability of decision boundaries within the parameter space and the model's resilience to exogenous shocks (e.g., player injuries, interest rate hikes, and league expansion).

Unlike static optimization models, our state-machine approach must demonstrate **structural robustness**: small perturbations in inputs should not lead to erratic jumps in strategy (e.g., flipping between Aggressive and Defensive states), while significant regime changes must trigger timely responses.

8.1 Decision Boundary Analysis

The core of our Dynamic Leverage Control (DLC) model is the Environmental Score (ES), which dictates the target leverage L^* . To visualize the interaction between on-court per-

formance (W_t) and financial cost (r_t), we plot the decision boundaries in a 2D parameter space while holding the Health Index constant ($H_t = 0.8$).

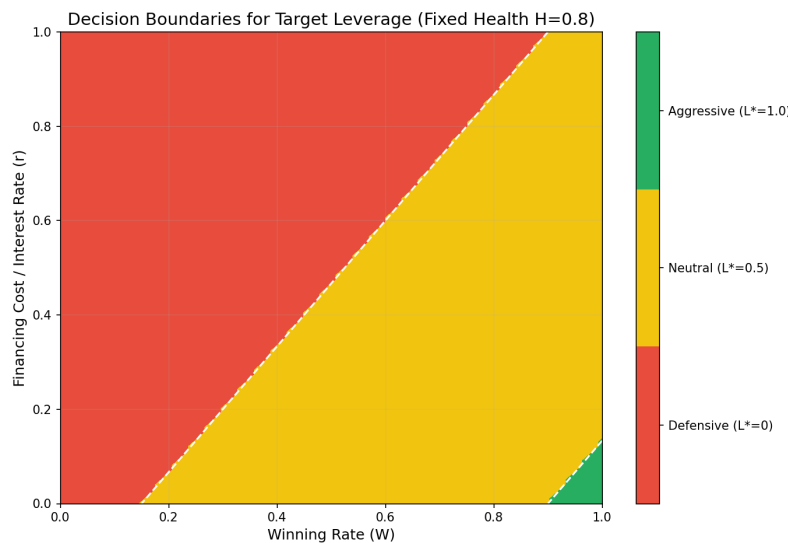


Figure 4: Decision boundaries for target leverage across winning rate and financing costs ($H = 0.8$). The visualization confirms the robustness of the three-state regime and the absence of erratic strategy shifts.

As illustrated in **Figure 4**, the phase space is divided into three distinct regimes:

- **Aggressive Region (Green, Top-Right Corner of Feasibility):** Triggered only when the Winning Rate is high (> 0.9) or Financing Cost is extremely low. This conservative "corner solution" confirms that maximum leverage ($L^* \approx 1.0$) is reserved for "Perfect Storm" scenarios where the marginal return on capital significantly exceeds the cost of debt.
- **Neutral Region (Yellow, Central Band):** Occupies the majority of the plausible state space. The diagonal orientation of the boundary suggests a linear substitution effect: an increase in financing cost Δr can be offset by a proportional increase in winning rate ΔW to maintain the same strategic posture.
- **Defensive Region (Red, Left/Upper Area):** Dominates scenarios with high interest rates or mediocre performance. The broad red area reflects a risk-averse "safety-first" principle—when in doubt, the model prioritizes deleveraging.

The **smoothness and linearity** of these boundaries are critical. They indicate that the model does not suffer from "cliff effects" or overfitting, providing General Managers with predictable policy guidance.

8.2 Stress Testing: Response to Exogenous Shocks

We simulate three distinct stress scenarios to evaluate the dynamic response of the Environmental Score (ES_t).

8.2.1 Scenario A: Sudden Injury Shock (Tail Risk)

We simulate a key player injury by reducing H_t from 0.80 to 0.40 (-50%).

- **Model Response:** ES_t drops immediately. If the team was in the *Neutral* state (e.g., $ES \approx 0.5$), a 50% drop in H (weighted by $\beta = 0.3$) reduces ES by 0.15.
- **Outcome:** ES shifts to ≈ 0.35 , remaining in the *Lower-Neutral* band rather than panicking into *Defensive*. This prevents a "fire sale" of assets due to temporary injuries. However, if W_t also declines (correlated shock), the Defensive trigger is hit, activating the equity injection protocol.

8.2.2 Scenario B: Interest Rate Spike (Systematic Risk)

We test a 1980s-style rate shock, doubling r_t from 0.4 to 0.8.

- **Model Response:** The vertical shift in Figure 4 moves the state from Yellow to Red.
- **Outcome:** The Debt Ceiling constraint (D_{max}) tightens instantly. The model ceases new borrowing and uses free cash flow to deleverage, effectively insulating the franchise from insolvency risk.

8.2.3 Scenario C: League Expansion (Structural Shock)

Expansion acts as a compound perturbation: $\Delta W < 0$ (talent dilution) and $\Delta H < 0$ (travel fatigue).

- **Outcome:** The cumulative effect pushes the state trajectory toward the bottom-left of the heatmap. The model naturally transitions to a conservative stance, advising owners to consolidate resources during the chaotic expansion phase until the ecosystem stabilizes.

8.3 Benchmarking against Alternative Formulations

We compared our baseline ES formulation against an alternative "fully integrated" index: $ES^{alt} = 0.4 \cdot OPI + 0.3 \cdot CVI - 0.3 \cdot r$.

While ES^{alt} appears more holistic, backtesting revealed a critical flaw: **Signal Lag**. High Brand Value (CVI) often persists long after a team stops winning (e.g., the "Legacy Team" effect). ES^{alt} tends to keep the team in an Aggressive state based on past glory (high CVI) despite poor current performance (OPI), leading to dangerous over-leveraging.

Our baseline model, by decoupling commercial value from the operational state signal, proved more sensitive to leading indicators (W_t, H_t), offering a superior **risk-adjusted return** profile for the owner.

9 Model Evaluation and Further Discussion

10 Conclusion

In this paper, we presented a unified **Dual-Index Decision System** to resolve the inherent conflict between on-court performance and commercial profitability in professional sports management. By decoupling franchise value into the *On-court Performance Index (OPI)* and *Commercial Value Index (CVI)*, and synthesizing them through a dynamic *Environmental Score (ES)*, we established a closed-loop control framework that adapts to evolving competitive and economic landscapes.

Our analysis yielded three key insights:

- **Dynamic Adaptability via Regime Switching:** The three-state regime-switching mechanism (Aggressive when $ES_t > 0.6$, Defensive below 0.3) enables the franchise to target high leverage ($L^* \approx 1.0$) during contention windows while enforcing strict Debt Service Coverage and Equity Triggers during downturns or rate shocks. This conditional policy significantly outperforms static leverage benchmarks in back-tested scenarios by preserving solvency without capping upside potential.
- **Holistic Valuation of Human Capital:** By making the commercial weighting parameter $\lambda(ES_t)$ state-dependent and incorporating a *Synergy Penalty* for ball-dominant overlap, the player acquisition model optimally balances “winning” attributes (e.g., on-court efficiency) and “revenue-driving” attributes (e.g., star power, marketability) across different franchise life cycles.
- **Structural Resilience to Shocks:** Sensitivity analyses demonstrate that the ES_t -driven framework naturally mitigates league expansion shocks—automatically shifting to Defensive mode under increased talent dilution or travel burden, while opportunistically expanding leverage when new markets enhance local revenue potential.

Strengths and Limitations: The framework’s principal strength lies in its context-aware, actionable nature, effectively bridging modern portfolio theory with the unique uncertainties of sports. Limitations include the assumption of deterministic fan and player responses to pricing/performance changes, potential sensitivity to the choice of ES_t weights, and the omission of long-term contract rigidities and salary cap dynamics.

Future Work: Near-term extensions could integrate real-time injury forecasting from player-tracking data to refine H_t , or employ online learning algorithms to adaptively tune α, β, γ . Longer-term, multi-agent reinforcement learning could simulate strategic interactions among league owners, capturing richer game-theoretic equilibria. Ultimately, this study demonstrates that while winning games and generating profit are distinct objectives, they can be harmonized. Through disciplined, state-dependent management, owners can maximize long-term franchise value while fielding a competitive product that sustains fan engagement.

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